VITALITY 05

Bremelanotide (PT-141)

Synthetic ring-shaped 7-amino-acid peptide modelled on alpha-melanocyte-stimulating hormone



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Peptide	Low desire in women, ED in men	Trials in women; small studies in men	FDA approved (Vyleesi, women); men off-label

STUDIED FOR

Low desire (women)	Erections	With sildenafil
HUMAN	HUMAN	HUMAN

One trial in men was flagged. The 2008 trial of 342 men from Iran has an expression of concern. Studies in men used a nasal spray or injection.

HOW IT WORKS IN THE BODY

Bremelanotide ▶ MC1R and MC4R ▶ Brain neurons ▶ Sexual desire

- **Activates melanocortin receptors;** MC1R and MC4R matter most (label)
- **MC4R** sits on nerve cells across the brain
- **MC1R** on skin pigment cells explains skin darkening
- **How it works** for low desire in women is unknown (label)

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Approved 2019 on 2 Phase III trials in women
- 1,267 women in the two 24-week trials
- Men: small studies, 2004 to 2008

NOT YET ESTABLISHED

- Side-effect profile in men
- Long-term use in men
- Oral, strip or troche forms

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
About 2.7 hours (range 1.9 to 4.0)	Injection under the skin (label); nasal spray in early studies in men
PHARMACOKINETICS	**SOLD AS**
Peak in about 1 hour; broken down by peptide hydrolysis	Prescription injection (Vyleesi); compounded forms unapproved
MONITORED IN TRIALS	**SPORT (WADA)**
Blood pressure, heart rate, skin darkening, nausea	Not on the 2026 list

REGULATORY

FDA	Approved 2019 for HSDD in premenopausal women
FDA	Label: not indicated in men
EMA	No EU-wide authorisation
WADA	Not on the 2026 list

REFERENCES

1. FDA, Vyleesi prescribing information (DailyMed), Nov 2025
2. Safarinejad and Hosseini, J Urol, bremelanotide in ED, Mar 2008
3. Diamond et al., Int J Impot Res, PT-141 nasal spray, Feb 2004

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